

Ivans Echo

A small solid-state echo box: one bucket-brigade line, narrow and hissy, from slapback to runaway.



Overview

Ivans Echo is in the spirit of the budget mini echo boxes of the early 80s: four knobs on a small metal box, with one bucket-brigade (BBD) chip inside. The input goes through a simple op-amp preamp, then a compander squeezes it into the chip and expands it again on the way out, so quiet passages breathe and the hiss sinks between notes. The echo path is narrow (about 4 kHz at the shortest delays, falling to about 2.6 kHz at the longest, because the chip's clock slows down) and it is mono, as on the original; the dry signal stays in stereo.

Short settings give a hard, metallic, obviously electronic sound; longer ones give a single slapback or dark, deep repeats. Turn REPEAT past about 2 o'clock and the echoes run away. As on a real BBD unit, turning DELAY TIME glides the delay rather than jumping, so the repeats bend in pitch while you turn it.

Reach for it for

- Rockabilly and rock'n'roll slapback
- Quick doubling (ADT) on vocals and guitars
- Metallic, boxy short echoes
- Lo-fi dub repeats and runaway feedback
- A cheap, characterful echo on a mic'd vocal

Controls

INPUT

INST / MIC	The two input sockets. INST is flat. MIC adds 12 dB of preamp gain, cuts the lows below about 150 Hz and brings up the hiss.
VOLUME	Input level into the preamp, -24 to +18 dB. Set it so the PEAK light only flickers; push it further and the preamp clips, roughly and a little unevenly.
PEAK	Lights as the preamp starts to clip.
FOOTSWITCH	Echo on or off (on by default). Off stops new signal reaching the echo, and the repeats already in it die away naturally.

ECHO

BALANCE	Dry to echo, 0 to 100% (35% by default). Up to 50% the dry signal stays at full level and the echo comes up; from 50% the dry fades out, leaving only the echo at 100%.
REPEAT	Feedback, 0 to 110% (35% by default): from a single echo to many. Past about 85% (2 o'clock) the repeats build up instead of fading; the loop clips like the original's op-amp, so it stays bounded.
REPEAT light	Shows the level in the feedback loop: it glows when the repeats are strong.
DELAY TIME	20 to 400 ms (180 ms by default). The longer the delay, the darker and grittier the echo. Changes glide, so the repeats bend in pitch as you turn it. With SYNC on, the knob is overridden and its readout shows the synced time.
SYNC	Click for a list: FREE, or a note value locked to the project tempo (1/32, 1/16T, 1/16, 1/8T, 1/16D, 1/8, 1/4T, 1/8D). The

time stays within 20 to 400 ms, so slow tempos may reach the limit.

MASTER

DRIFT	See Drift below.
OUTPUT	The output level, -24 to +12 dB.
OUT meters	Output level, left and right. Click them to reset the peak hold.

The header

PRESETS	Opens the list of factory presets (or click the preset display). Choosing one sets everything except FOOTSWITCH and OUTPUT.
◀ / ▶	Step through the presets.
Preset display	Shows the preset number and name. Click it for the list.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

Ivans Echo has no input guard and no output stage: the levels are set with VOLUME and OUTPUT only.

Drift

DRIFT (0 - 100%, 50% by default) works on two things. The chip's clock wanders, so the delay time breathes a little. And the supply voltage wanders and sags: the preamp's headroom and gain move slowly, and loud passages pull the supply down so the preamp squashes a little more. 50% is the standard amount, 100% twice that, 0 steady; the wander also speeds up as you turn it up.

Presets

Init	The default settings: INST, 180 ms, 35% repeat.
Slapback	110 ms, a single repeat, the input pushed a little.
ADT Double	28 ms, no repeats, more drift: a quick double.
Metallic Room	24 ms with heavy repeats: the hard, metallic short-echo sound.
Dub Repeats	375 ms, 68% repeat: dark, long repeats.
Runaway	260 ms, 98% repeat, a hot input and lots of drift: the echoes build up on their own.
Mic Booth	MIC input, 140 ms, a few repeats.
Lo-Fi Haze	320 ms, a clipped input and maximum drift.

Tips

- For dub throws, tap FOOTSWITCH off and on to catch single words, and ride DELAY TIME by hand for pitch swoops.
- On a send, set BALANCE to 100% so only the echo comes back.
- If the repeats get muddy, shorten the delay: the echo opens up as the clock speeds up.
- Use MIC for a rougher, hissier character, even on line-level sources (turn VOLUME down to suit).

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